

Software Engineering LAB-01

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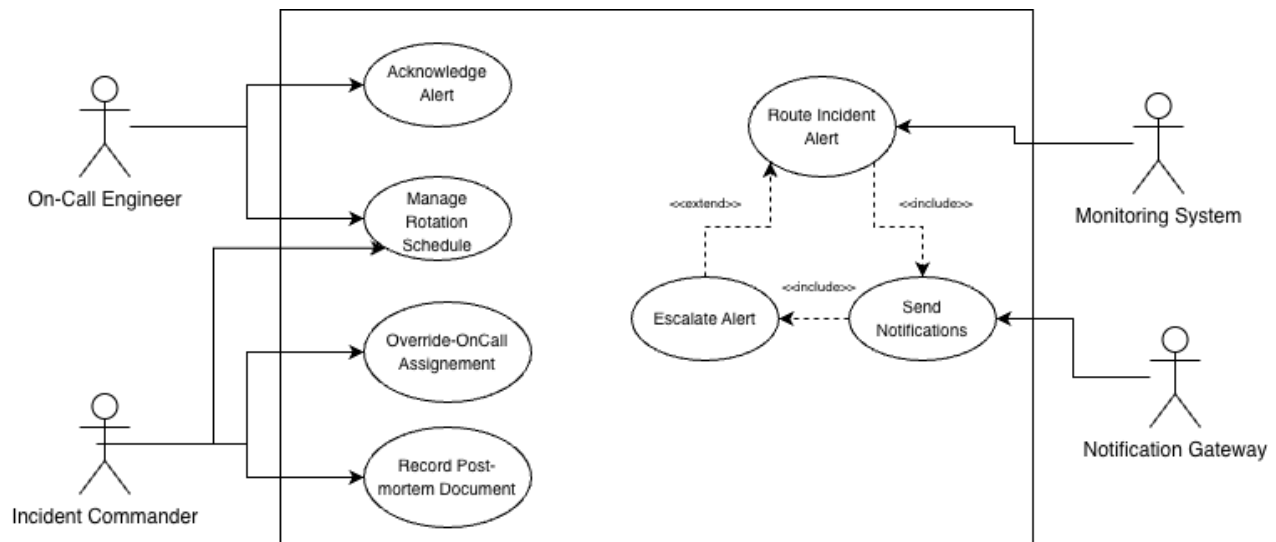
Problem Statement Number: 48

Functional/Non-Functional Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall take incident alert, identify the active on-call engineer from the rotation schedule, and then can escalate to the secondary engineer if it is unacknowledged within 5 minutes.	High	Pass: If the escalation step is triggered before the timeout Fail: If the unacknowledged P1 is dropped without escalation	Ensures no critical alert is missed silently
FR-002	Functional	The system shall allow the on-call engineer to acknowledge an alert via SMS reply, Outlook which stops further escalation for that incident	High	Pass: Acknowledging within the escalation window, halts the timer and no further notifications are sent. Fail: Not able to acknowledge within the window, thus escalation continues	Prevents unnecessary noise once someone is responding
FR-003	Functional	The system shall allow an admin to create and edit rotation schedules including assigned engineers, on-shift/off-shift timings	Medium	Pass: A new/updated schedule correctly determines on-call engineers at any required timestamps Fail: No on-call engineers returned for there valid on-shift timings within a defined schedule	Rotation accuracy is the foundation the whole escalation logic depends on.

FR-004	Functional	The system shall allow Incident Commander to manually override the current assigned on-call engineer for an active or upcoming shift	Medium	Pass: Override is applied, and the updated alerts now route to the new engineer Fail: Override applied, but the updated alerts are routed to the previously assigned engineer	It gives the team to handle unplanned changes like sick days or shift swaps without editing the whole schedule
FR-005	Functional	The system should allow the Incident Commander to document a resolved incident, including timeline root cause, and action items	Low	Pass: A saved-document can be retrieved for the originating incident Fail: The document is not saved or is not linked to the correct originating incident	Supports learning/process improvement after incidents.
NFR-001	Non-Functional	(Performance & Security)Alert dispatch via webhook, SMS, and email must initiate within 3 seconds of alert ingestion.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Delay in initial notification directly delays incident response.
NFR-002	Non-Functional	The system's alert ingestion and escalation service shall maintain 99.9% uptime measured monthly.	High	Pass: Uptime monitoring logs show $\geq 99.9\%$ availability over a rolling 30-day window. Fail: Downtime exceeds ~43 minutes/month, causing missed alert routing.	The escalation engine is only useful if it's reliably running when incidents happen.

UML Diagram



Use Case: Route & Escalate Incident Alert

Preconditions:

- A monitoring system is configured to send alerts to the escalation engine via sms alerts or outlook.
- An active on-call rotation schedule exists with at least a primary and secondary engineer assigned.

Postconditions:

- The incident alert has been acknowledged by an on-call engineer, and the incident is marked "acknowledged" with a timestamp and owner.

Main Success Scenario:

1. Monitoring system sends an incident alert to the system via webhook.
2. System looks up the active on-call rotation schedule and identifies the primary on-call engineer.
3. System dispatches a notification (SMS, email, webhook) to the primary engineer.
4. Primary engineer receives the notification and acknowledges the alert within the app.
5. System records the acknowledgment (engineer ID, timestamp) and stops the escalation timer.
6. Use case ends successfully.

Alternate Flow

4a. Alert Unacknowledged Within 5 Minutes

- System detects that 5 minutes have elapsed with no acknowledgment.
- System escalates the alert to the secondary engineer and dispatches a new notification.
- If the secondary engineer also fails to acknowledge within 5 minutes, the system escalates to the Incident Commander and flags the incident as "critical: unacknowledged."